



Robots in the Lab!

Dəyərli qrup üzvləri
 Dəyərli qrup üzvləri! Əməkdaşlıq
 5423; 25358



Overview

- Background
- What is Automation
- Automation Examples
- Projects




Education

BEng - Bioengineering
 • Yildiz Technical University - Istanbul/Turkey

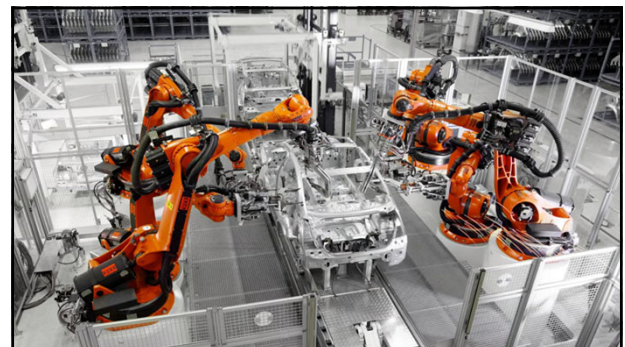
MSc - Applied Biopharmaceutical Biotechnology and Entrepreneurship
 • University of Nottingham - Nottingham/UK





What is automation?

Dəyərli qrup üzvləri! Əməkdaşlıq
 z. 14.10.19. Dəyərli qrup üzvləri!

What is Lab Automation?



What is Lab Automation?



What is Lab Automation?



What is Lab Automation?



What is Lab Automation?

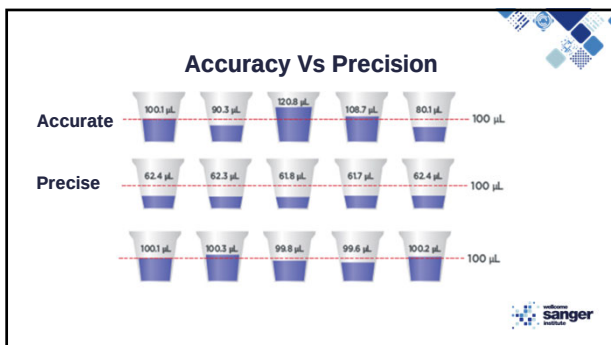
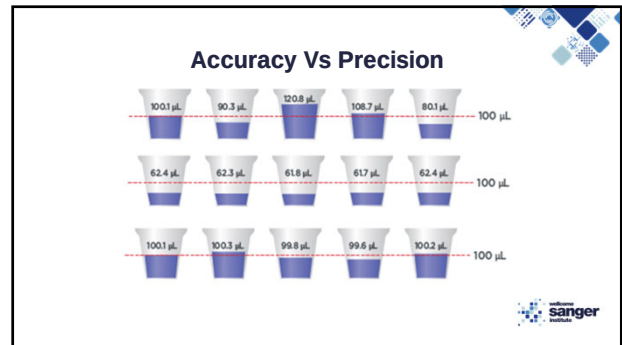


What is Lab Automation?



Why?

1. High throughput and accuracy
1. Reduces human error
1. Improves safety and hygiene
1. Saves time and resources

Examples of instruments used in Lab Automation



Hamilton

[i3ezi3y er)3j3diwi3eq p3erw3r\$md2X11)33v33rs{ r\$
j3v3di3m3v3gn3r33rh33 sh3yv3v3\$3gerf3i333dk3v3dh3\$ m3\$
s3di3v3r3w3y3q irw-2
M333w3\$33r3w33rh33sq iw3\$ m3333i3eh3\$ m33333v33<8\$
gl3rrip3\$

1. Hamilton Pooling

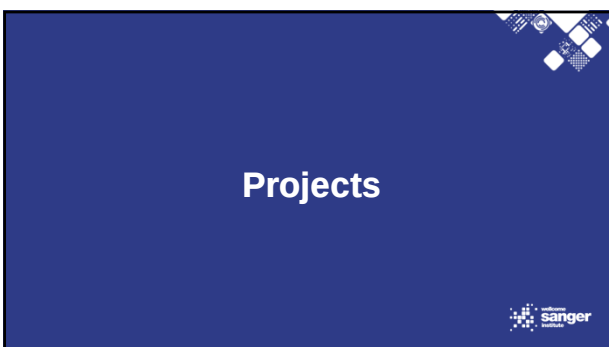
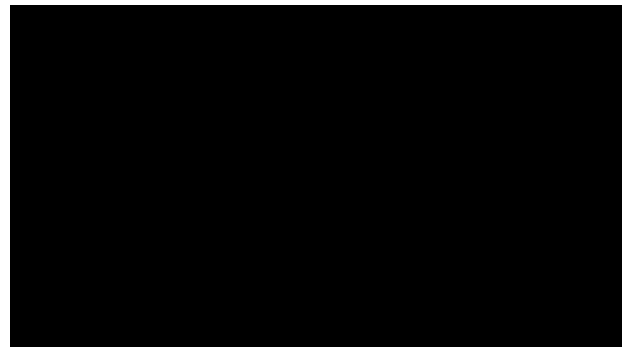
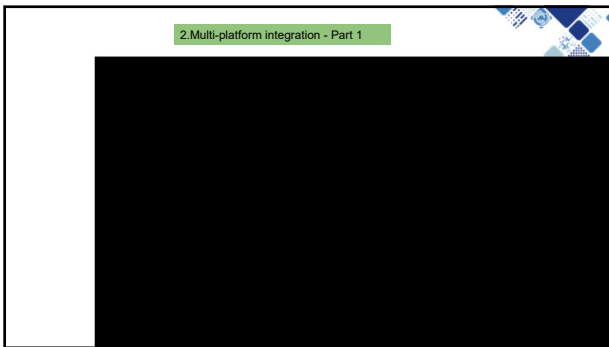
To access the videos, use this link:
<https://drive.google.com/drive/folders/18yzqTgsH5S5YT-M0 - Q04Tm5pFWYcG>



Hamilton

[i3ezi3y er)3j3diwi3eq p3erw3r\$md2X11)33v33rs{ r\$
j3v3di3m3v3gn3r33rh33 sh3yv3v3\$3gerf3i333dk3v3dh3\$ m3\$
s3di3v3r3w3y3q irw-2
M333w3\$33r3w33rh33sq iw3\$ m3333i3eh3\$ m33333v33<8\$
gl3rrip3\$





Bioscan

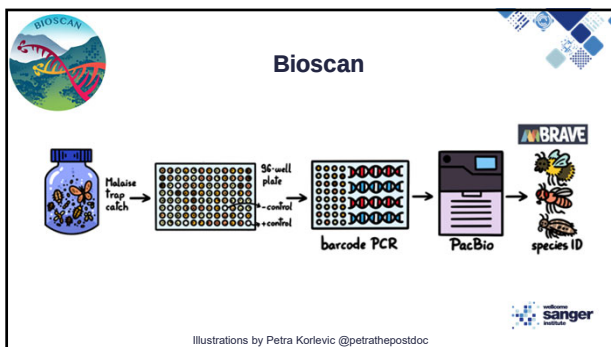
0 E 2 y f d g # u r n f # i p v i e # h t x d g f n # # p h k q # d j i j # i y v h f w # i f u r v v # k h k N # g j B #
| h d u 2

0 V d p s d v h t q # i x p # d u e g h u # h v # i f u r v v # k h k N z k h u # k h j # d e w h k h #
i y v h f w # v j j # d a l h h # d e v 1





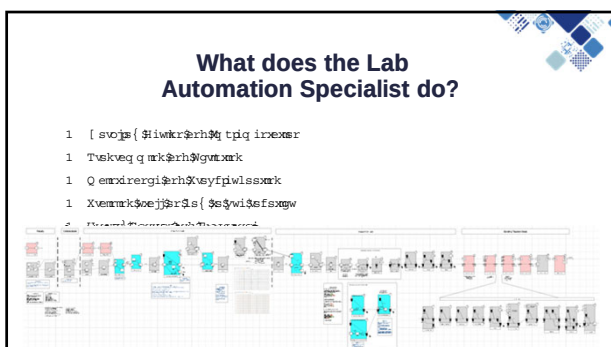
sanger

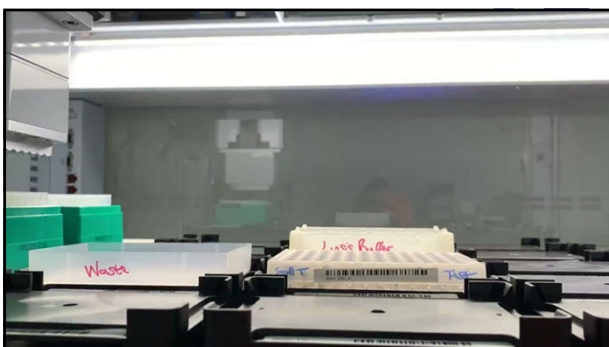
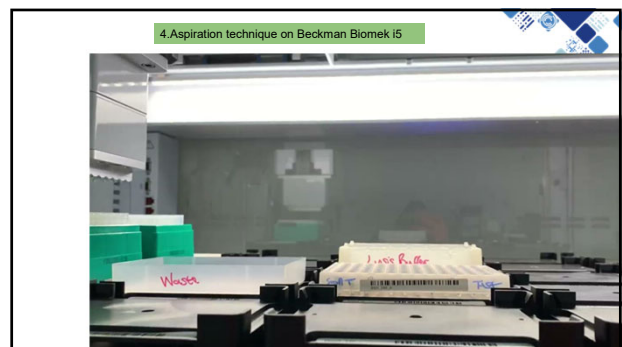
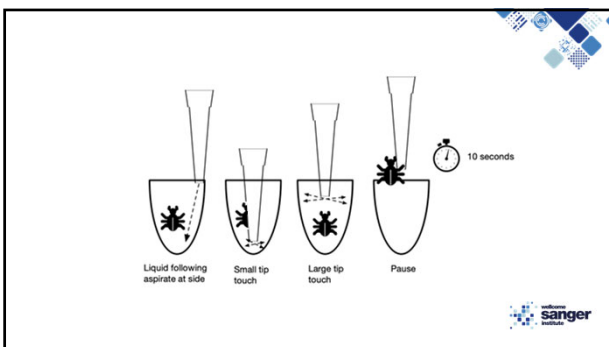
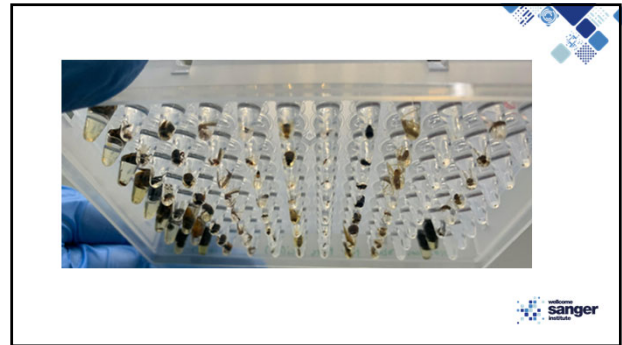
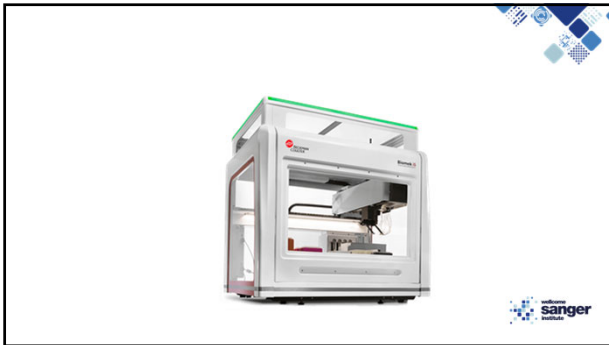


What does the Lab Automation Specialist do?

- 1 [s v o j s (# i w k r # e r h # g t p i q i n e x m r
- 1 T e k v e q q m k # e r h # g u t x a k
- 1 Q e m d r e r g i # e r h # v e y f p w l s s a x k
- 1 X v e m k # o e j j # e r s (# s # w i s e s x g w
- 1 U y e m # s e r x e p # e r h # w w y e r g i

sanger





Programing and Scripting

```

test_programs() {
  for i in $(seq 1 10); do
    echo "Running test $i"
    # Run the test program
    python3 test_program.py $i
    # Check the results
    if [ $? -eq 0 ]; then
      echo "Test $i passed"
    else
      echo "Test $i failed"
    fi
  done
}

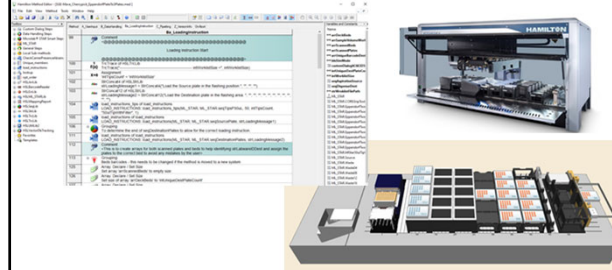
# Run the tests
test_programs

```

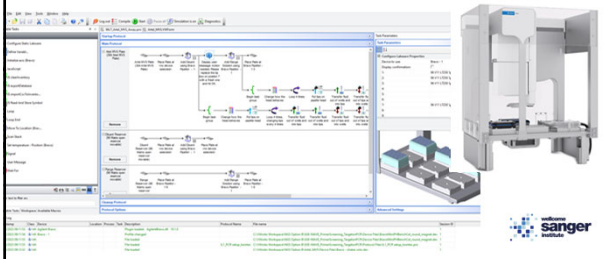


sanger

Programing and Scripting



Programing and Scripting



Other Automation Examples

sanger



YT Links

HTKv4b4e-[sanger@sanger.ac.uk](#)
Tread[4R] [arg4]dprach4Rf-[sanger@sanger.ac.uk](#)
Uryu4R [arg4]dprach4Rf-[sanger@sanger.ac.uk](#)
Rrpo4R-[sanger@sanger.ac.uk](#)



Thanks for listening!

